

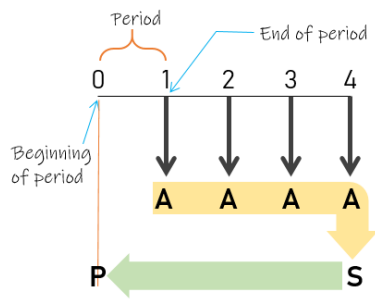
Annuity A uniform series of payments occurring at equal interval of time.

Types of Annuity

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1. ORDINARY ANNUITY

A type of annuity where the payments are made at the end of each period starting from the first period.



Sum:

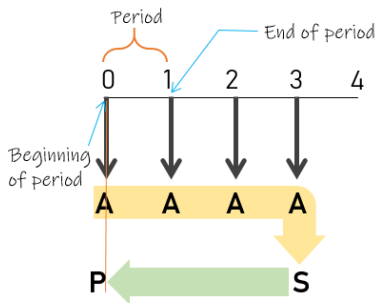
$$S = \frac{A[(1+i)^n - 1]}{i}$$

Present worth:

$$P = \frac{A[(1+i)^n - 1]}{i(1+i)^n}$$

2. ANNUITY DUE

A type of annuity where the payments are made at the beginning of each period starting from the first period.

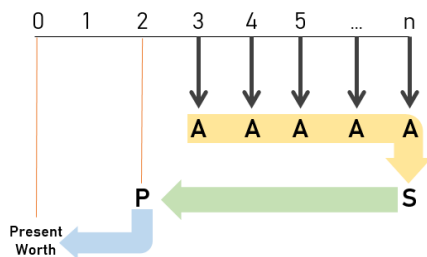


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Present Worth = $P + A$

3. DEFERRED ANNUITY

A type of annuity that does not start immediately on the beginning of each period

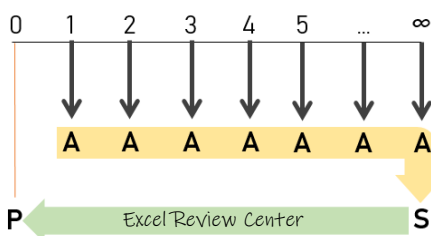


For present worth,
Use compound
interest formula

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4. PERPETUITY

An ordinary annuity that continues indefinitely or forever.



Sum of perpetuity
is always equal to
infinity

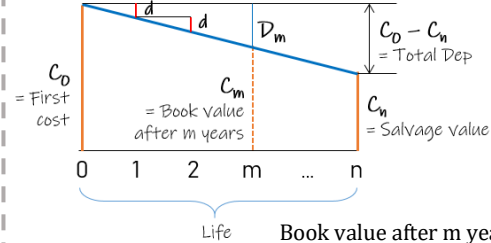
$$P = \frac{A}{i}$$

Depreciation The reduction or fall of the value of an asset due to constant use and passage of time.

Methods of Computing Depreciation

1. STRAIGHT LINE METHOD

Annual dep. charge



$$d = \frac{C_0 - C_n}{n}$$

Total dep after m years

$$D_m = d(m)$$

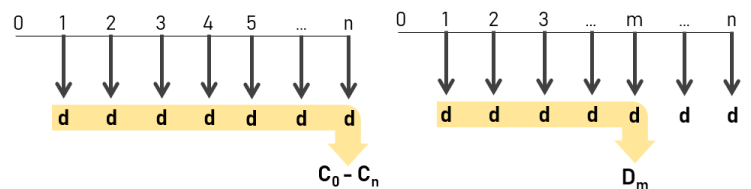
$$C_m = C_0 - D_m$$



First cost – purchased price + all expenses to make equipment operational
Salvage value – also called Second-hand value
Scrap value – also known as Junk value

2. SINKING FUND METHOD

In this method, interest is included in computing the depreciation.



Annual dep. charge

$$d = \frac{(C_0 - C_n)i}{[(1+i)^n - 1]}$$

Total dep. after m years

$$D_m = \frac{d[(1+i)^m - 1]}{i}$$

Book value after m years

$$C_m = C_0 - D_m$$

3. DECLINING BALANCE METHOD

Also known as Constant Percentage Method

Matheson Formula

$$k = 1 - \sqrt[n]{\frac{C_n}{C_0}}$$

$$k = 1 - \sqrt[m]{\frac{C_m}{C_0}}$$



This method cannot have a salvage value of zero!

k = constant percentage

4. SUM-OF-YEARS DIGIT (SYD) METHOD

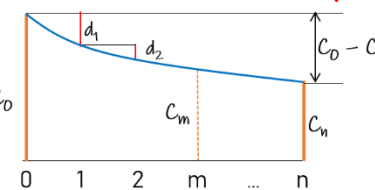
Corresponding dep charges:

$$d_1 = (C_0 - C_n) \frac{n}{\sum \text{year}}$$

$$d_2 = (C_0 - C_n) \frac{n-1}{\sum \text{year}}$$

$$d_3 = (C_0 - C_n) \frac{n-2}{\sum \text{year}}$$

... and so on.



Sum of year:

$$\sum \text{year} = \frac{n(n-1)}{2}$$

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